Quick start for LaTeXing with the IEEEtran class for IEEE Computer Society Conferences

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***Abstract*—Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.**

I. Introduction

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metu enim. Vestibulum pellentesque felis eu massa.TODO!

The remainder of the paper starts with a presentation of related work (Section II). It is followed by a presentation of hints on LaTeX (Section IV). The paper closes with a conclusion and an outlook on future work (Section V).

II. Related W

Winery [1] is a graphical modeling tool. The whole idea of TOSCA is explained by Binz et al. [2].

Tools for the literature search are, among others, ScienceOS, dblp, Google Scholar, IEEE Xplore, and the ACM Digital Library; JabRef manages the found papers, fetches the bibliographic data by DOI, and keeps the PDFs. If the search is systematic (as in a literature review), the keywords, the time span, and the inclusion criteria belong into the thesis so that a reader can repeat the search. Writing hints for student theses are given in the open-access book *Student Thesis Projects*[1]. More hints are given in *Writing for Computer Science*[2].

III. Writing in Markdown

This section uses plain Markdown, not LaTeX. Its source is the file `content-markdown.md`, which the main file includes with the `markdownInput` command of the `markdown` package[3]. This needs LuaLaTeX. Text can be *emphasized* or **bold**, and a citation looks like this [3]. Section IV-O shows more Markdown.

IV. LaTeX Hints

This section contains hints on writing LaTeX. It focuses on minimal examples, which can be directly adapted to the content

[1] https://vdf.ch/product/student-thesis-projects-en.html
[2] https://doi.org/10.1007/978-1-4471-6639-9
[3] https://ctan.org/pkg/markdown

A. *Handling of paragraphs*

One sentence per line. This rule is important for the usage of version control systems. A new line is generated with a blank line. As you would do in Word: New paragraphs are generated by pressing enter. In LaTeX, this does not lead to a new paragraph as LaTeX joins subsequent lines. To start a new paragraph, press enter twice. This leads to an empty line. In word, there is the functionality to press shift and enter. This leads to a hard line break. The text starts at the beginning of a new line. In LaTeX, you can do that by using two backslashes (\\).
Running text seldom needs a hard line break.

Please do *not* use two backslashes for new paragraphs. For instance, this sentence belongs to the same paragraph, whereas the last one started a new one. Andrew Stacey gives a long motivation for that in the section "Line Wrapping" of his *Document Revision System*[4].

Corresponding LaTeX code of `paper-basic-listings.tex`

```
868 œŒŭŐŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
869 One sentence per line.
870 This rule is important for the usage of
        ↪ version control systems.
871 A new line is generated with a blank line.
872 As you would do in Word:
873 New paragraphs are generated by pressing
        ↪ enter.
874 In LaTeX, this does not lead to a new
        ↪ paragraph as LaTeX joins subsequent
        ↪ lines.
875 To start a new paragraph, press enter twice
        ↪ .
876 This leads to an empty line.
877 In word, there is the functionality to
        ↪ press shift and enter.
878 This leads to a hard line break.
879 The text starts at the beginning of a new
        ↪ line.
880 In LaTeX, you can do that by using two
        ↪ backslashes (\textbackslash\
        ↪ textbackslash).
881 \\
882 Running text seldom needs a hard line break
        ↪ .
883
884 Please do \textit{not} use two backslashes
        ↪ for new paragraphs.
885 For instance, this sentence belongs to the
        ↪ same paragraph, whereas the last one
        ↪ started a new one.
886 Andrew Stacey gives a long motivation for
        ↪ that in the section \enquote{Line
        ↪ Wrapping} of his \href{https://
        ↪ loopspace.mathforge.org/HowDidIDoThat
        ↪ /TeX/VCS/#section.3}{\emph{Document
        ↪ Revision System}}\footnote{\url{https
        ↪ ://loopspace.mathforge.org/
        ↪ HowDidIDoThat/TeX/VCS/}}.
```

Once the text is final, single paragraphs can be polished. If the last line of a paragraph holds only a word or two, or is nearly full, let TeX set the paragraph one line shorter or longer: write `\looseness=-1` (or `=1`) as the first thing in that paragraph. TeX resets `\looseness` at the end of every paragraph, so only that one paragraph is affected. With LuaLaTeX, a lone short word as the last line is avoided automatically (package impnattypo).

[4]https://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/

B. *Notes separated from the text*

The package mindflow enables writing down notes and annotations in a way so that they are separated from the main text.

This is a small note.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
898 œŒŭŐŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
899 \begin{mindflow}
900 This is a small note.
901 \end{mindflow}
```

C. *Handling TODOs*

Marked text.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
906 œŒŭŐŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
907 \textmarker{Marked text.}
```

With `\textmarker`, only the text color is changed, since this also works well for individual words.
Marked text.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
912 œŒŭŐŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
913 \textcomment{Marked text.}{A comment on it
        ↪ .}
```

Manual marking for text that has been changed since the last version.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
916 œŒŭŐŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
917 \modified{Manual marking for text that has
        ↪ been changed since the last version.}
```

This is some text. Changed text.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
This is some text.
\change{FL1: text adjusted}{Changed text}.
```

Here is a comment.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
Here is a comment\sidecomment{Comment}.
```

TODO!

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
\todo{A lot of text still needs to be
    ↪ produced here}
```

Several authors can mark their edits and comments with the phfcc package; the initials of the author appear once per page in the margin. Define one command per author in the preamble (`\phfMakeCommentingCommand`); `\phfDisableCommentingCommands` hides everything for the final version.

Quantum mechanics developed quickly thanks to the works of Schrödinger, Heisenberg, and Dirac, who drew inspiration from earlier ideas by Planck FA and Einstein [Einstein must not be missing here.] ~~This sentence is to be removed.~~ **!!! This is important. !!!**

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
Quantum mechanics developed quickly thanks
    ↪ to the works of Schrödinger,
    ↪ Heisenberg, and Dirac,
who drew inspiration from earlier ideas by
    ↪ Planck \FA{and Einstein} \FA[Einstein
    ↪  must not be missing here.]
\FA*{This sentence is to be removed.} \FA!{
    ↪ This is important.}
```

D. Hyphenation

LaTeX automatically hyphenates words. When using microtype, there should be fewer hyphenations than in other settings. You might still need to tweak the hyphenation. Here are some hints:

In case you write “application-specific”, then the word will only be hyphenated at the dash. You can also insert `\allowbreak{}` into the word, as in application-specific, but this is much more effort.

You can now write words containing hyphens which are hyphenated at other places in the word. For that, write `"=` instead of −, as in application”=specific. This is enabled by an additional configuration of the babel package.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
In case you write \enquote{application-
    ↪ specific}, then the word will only be
    ↪ hyphenated at the dash.
You can also insert \verb1\allowbreak{}1
    ↪ into the word, as in applica\
    ↪ allowbreak{}tion-specific, but this is
    ↪  much more effort.

You can now write words containing hyphens
    ↪ which are hyphenated at other places
    ↪ in the word.
For that, write \verb1"=1 instead of \verb1
    ↪ -1, as in application"=specific. %
    ↪ chktex 18
This is enabled by an additional
    ↪ configuration of the babel package.
```

E. Typesetting Units

Numbers can be written plain text (such as 100), by using the siunitx package as follows: 100 $\frac{\mathrm{km}}{\mathrm{h}}$, or by using plain LaTeX (and math mode): 100$\frac{km}{h}$.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
Numbers can be written plain text (such as
    ↪ 100), by using the \href{https://ctan
    ↪ .org/pkg/siunitx}{siunitx} package as
    ↪  follows:
\SI{100}{\km\per\hour},
or by using plain \LaTeX{} (and math mode):
$100 \frac{\mathit{km}}{h}$.
```

5 % of 10 kg

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
\SI{5}{\percent} of \SI{10}{kg}
```

Numbers are automatically grouped: 123 456.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
Numbers are automatically grouped: \num{123
    ↪ 456}.
```

F. Surrounding Text by Quotes

Please use the “enquote command” to quote something. Quoting with “quote” or “quote” also works.

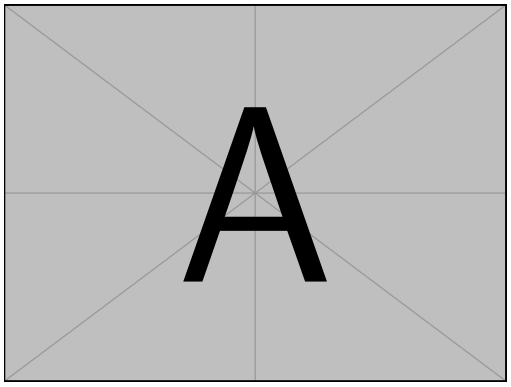


Fig. 1. Example figure for zcref demo

Table I
EXAMPLE TABLE FOR ZCREF DEMO

Heading1	Heading2
One	Two
Three	Four

Corresponding LaTeX code of paper-basic-listings.tex

```
977 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
978 Please use the \enquote{enquote command} to
        ↪  quote something.
979 Quoting with "`quote"' or ``quote'' also
        ↪ works. % chktex 18
```

G. Zref-clever examples

Demonstration of zref-clever: Zcref (short for zcref[S]) at the beginning of a sentence, zcref in all other cases.

Fig. 1 shows a simple fact, although Fig. 1 could also show something else.

Table I shows a simple fact, although Table I could also show something else.

Section IV-G shows a simple fact, although Section IV-G could also show something else.

Corresponding LaTeX code of paper-basic-listings.tex

```
1009 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
1010 \Zcref{fig:ex:cref} shows a simple fact,
        ↪ although \zcref{fig:ex:cref} could
        ↪ also show something else.
1011
1012 \Zcref{tab:ex:cref} shows a simple fact,
        ↪ although \zcref{tab:ex:cref} could
        ↪ also show something else.
1013
1014 \Zcref{sec:ex:cref} shows a simple fact,
        ↪ although \zcref{sec:ex:cref} could
        ↪ also show something else.
```

H. Figures

Fig. 2 shows something interesting.

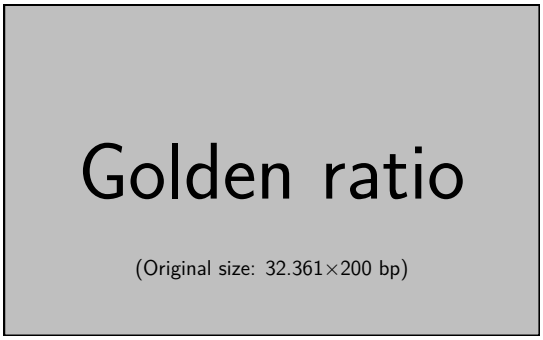


Fig. 2. Simple Figure. Based on Scharrer [3].



Fig. 3. Ulm Minster. Photo: wkrauth, CC0, via Wikimedia Commons.

Corresponding LaTeX code of paper-basic-listings.tex

```
1019 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
1020 \Zcref{fig:label} shows something
        ↪ interesting.
1021
1022 \begin{figure}
1023   \centering
1024   \includegraphics[width=.8\linewidth]{
          ↪ example-image-golden}
1025   \caption[Simple Figure]{
1026     Simple Figure.
1027     Based on \citet{mwe}.
1028   }
1029   \label{fig:label}
1030 \end{figure}
```

Photos in JPEG or PNG format are included the same way, as shown in Fig. 3.

Corresponding LaTeX code of paper-basic-listings.tex

```
1035 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
1036 \begin{figure}
1037   \centering
1038   \includegraphics[width=.5\linewidth]{
          ↪ figures/ulm-minster.jpg}
```

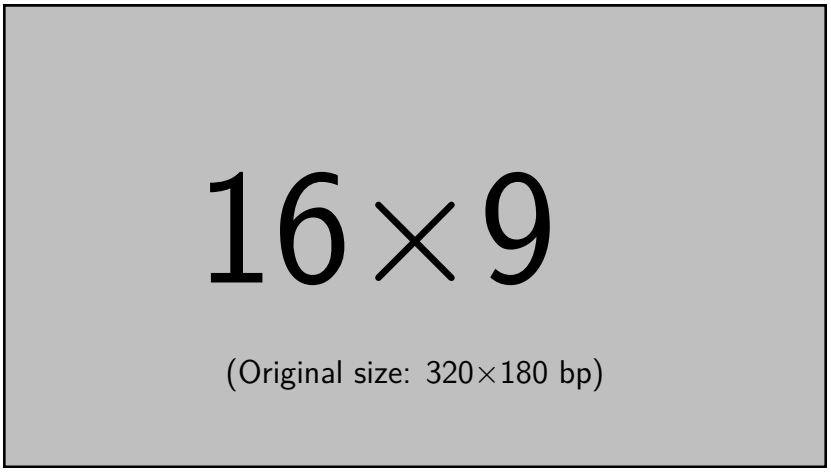


Fig. 4. Figure in 16:9 format

```
  \caption{Ulm Minster. Photo: wkrauth, CC0
      ↪ , via Wikimedia Commons.}
  \label{fig:jpg}
\end{figure}
```

A figure spans multiple columns when using `\begin{figure*}`. See Fig. 4 as an example.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒŰŐŐŁłıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{figure*}
  \centering
  % note that \textwidth is used instead of
      ↪  \linewidth
  % This ensures that the graphics width is
      ↪  60% of the "page" (text block),
      ↪ and not just 60% of the current
      ↪ text column
  % See https://tex.stackexchange.com/a
      ↪ /17085/9075 for details
  \includegraphics[width=.6\textwidth]{
      ↪ example-image-16x9}
  \caption{Figure in 16:9 format}
  \label{fig:16x9}
\end{figure*}
```

I. Sub Figures

An example of two sub figures is shown in Fig. 5.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒŰŐŐŁłıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{figure*}[!t]
  \centering
  \subfloat[Case I]{\includegraphics[width=.
      ↪ 4\linewidth]{example-image-a}%
    \label{fig:first_case}}
  \hfil
  \subfloat[Case II]{\includegraphics[width
      ↪ =.4\linewidth]{example-image-b}%
    \label{fig:second_case}}
  \caption{Example figure with two sub
      ↪ figures.}
  \label{fig:two_sub_figures}
\end{figure*}
```

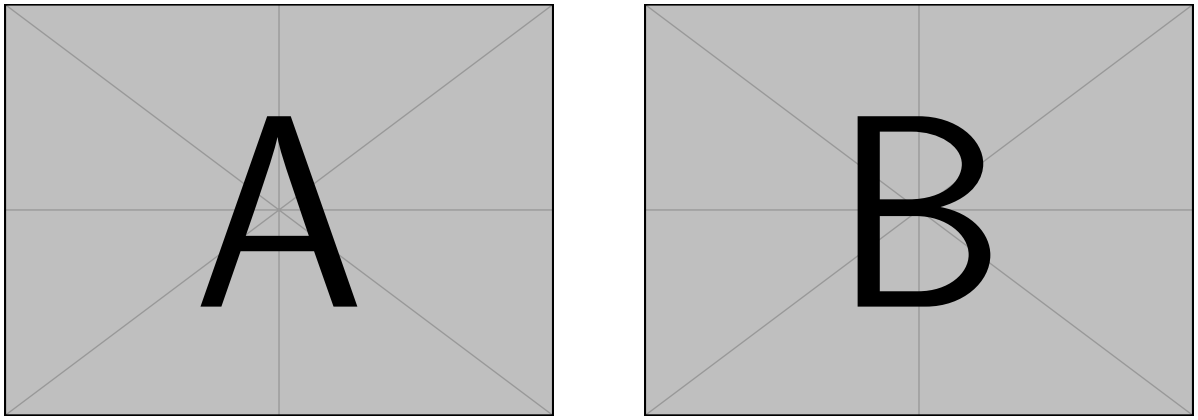
Note that often IEEE papers with subfigures do not employ subfigure captions (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for the subfig package to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g., `\subfloat[]`. An example is shown in Fig. 6.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒŰŐŐŁłıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{figure*}[!t]
  \centering
  \subfloat[]{\includegraphics[width=.4\
      ↪ linewidth]{example-image-a}%
    \label{fig:first_case_ieee}}
  \hfil
  \subfloat[]{\includegraphics[width=.4\
      ↪ linewidth]{example-image-b}%
    \label{fig:second_case_ieee}}
  \caption{Example figure with two sub
      ↪ figures in IEEE style: (a) the first
      ↪ case, (b) the second case.}
  \label{fig:two_sub_figures_ieee}
\end{figure*}
```

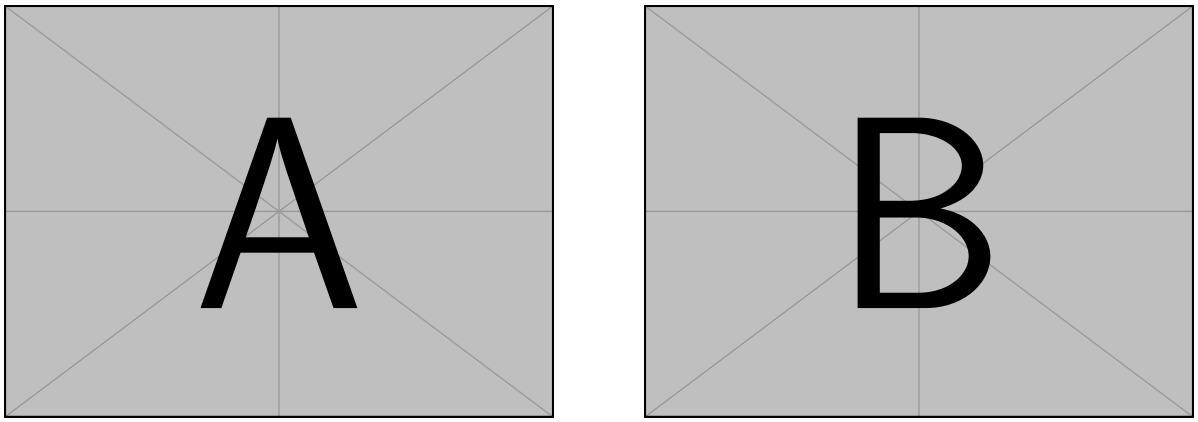
J. Figures with tikz

The tikz package creates graphics programmatically. This package generates grids and other regular structures.



(a) Case I (b) Case II

Fig. 5. Example figure with two sub figures.



(a) (b)

Fig. 6. Example figure with two sub figures in IEEE style: (a) the first case, (b) the second case.

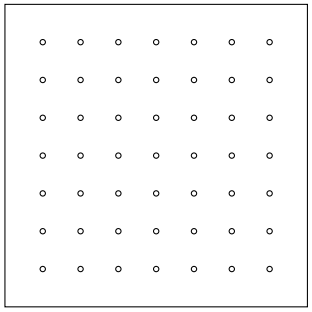


Fig. 7. A regular grid generated easily with two for loops.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
1104 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
1105 \begin{figure}[ht]
1106   \centering
1107   \begin{tikzpicture}
1108     \draw(0,0) rectangle (4,4);
1109     \foreach \x in {0.5,1,1.5,2,2.5,3,3.5}
1110     \foreach \y in {0.5,1,1.5,2,2.5,3,3.5}
1111     \draw(\x,\y) circle (1pt);
1112   \end{tikzpicture}
1113   \caption{A regular grid generated easily
          ↪ with two for loops.}\label{fig:tikz_
          ↪ example}
1114 \end{figure}
```

K. Plots with pgfplots

The pgfplots package plots functions directly in LaTeX, similar to MATLAB or gnuplot. Some visual examples are available in the package documentation[5].

Corresponding LaTeX code of `paper-basic-listings.tex`

```
1122 œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
1123 \begin{figure}[ht]
1124   \centering
```

[5]http://texdoc.net/pkg/pgfplots

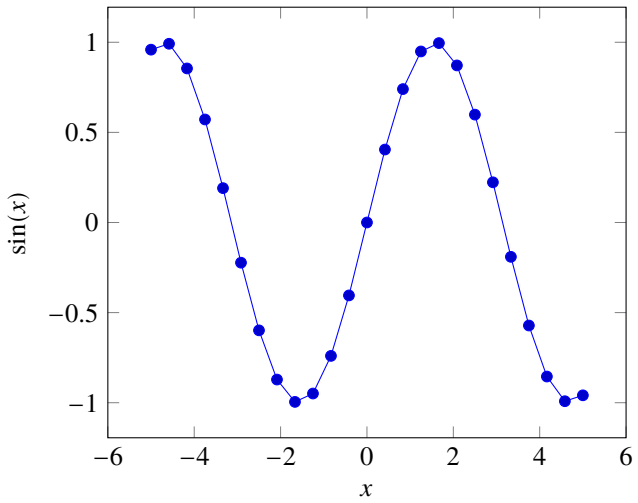


Fig. 8. Plot of $\sin(x)$ directly inside the figure environment with pgfplots.

```
\begin{tikzpicture}
  \begin{axis}[xlabel=$x$, ylabel=$\sin(x)$]
    \addplot {sin(deg(x))}; % Plot the sine function
  \end{axis}
\end{tikzpicture}
\caption{Plot of $\sin(x)$ directly inside the figure environment with pgfplots.}
\label{fig:pgfplots_example}
\end{figure}
```

Coordinates can also be given explicitly instead of as a function:

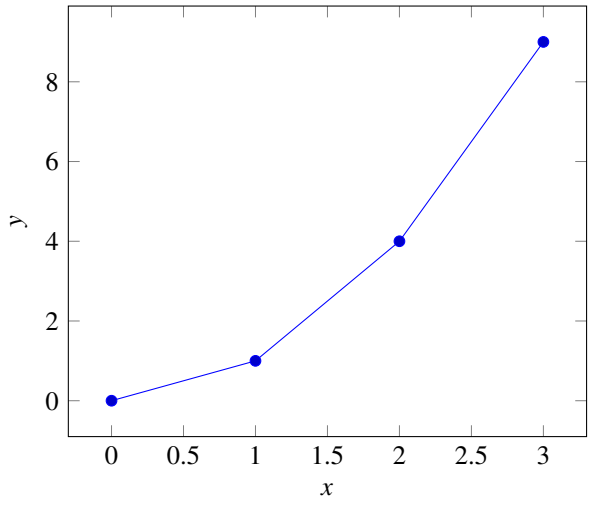


Fig. 9. Explicit coordinates plotted with pgfplots.

Table II
SIMPLE TABLE

Heading1	Heading2
One	Two
Three	Four

Table III
TABLE WITH COLORED CELLS

Test	Result
A	pass
B	fail

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
\begin{figure}[ht]
  \centering
  \begin{tikzpicture}
    \begin{axis}[xlabel=$x$, ylabel=$y$]
      \addplot coordinates {(0,0) (1,1) (2,4) (3,9)}; % Plot explicit coordinates
    \end{axis}
  \end{tikzpicture}
  \caption{Explicit coordinates plotted with pgfplots.}
  \label{fig:pgfplots_coords}
\end{figure}
```

L. Tables

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
\begin{table}
  \caption{Simple Table}
  \label{tab:simple}
  \centering
  \begin{tabular}{ll}
    \toprule
    Heading1 & Heading2 \\
    \midrule
    One      & Two      \\
    Three    & Four     \\
    \bottomrule
  \end{tabular}
\end{table}
```

A table caption goes above the table (`\caption` before the `tabular`), a figure caption below the figure.

Rows and cells can be colored with `\rowcolor` and `\cellcolor` from the colortbl package, which xcolor loads with its option `table`.

Table IV
TABLE WITH A FOOTNOTE

Heading1	Heading2[6]
One	Two

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
\begin{table}
  \caption{Table with colored cells}
  \label{tab:colored}
  \centering
  \begin{tabular}{ll}
    \toprule
    \rowcolor{gray!20}
    Test & Result              \\
    \midrule
    A    & \cellcolor{green!20}pass \\
    B    & \cellcolor{red!20}fail \\
    \bottomrule
  \end{tabular}
\end{table}
```

A `\footnote` inside a floating table is lost. Use `\tablefootnote` from the tablefootnote package instead: it is set as a normal footnote at the bottom of the page. If the notes should appear directly below the table instead (common in journals), use the threeparttable package.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
\begin{table}
  \caption{Table with a footnote}
  \label{tab:footnote}
  \centering
  \begin{tabular}{ll}
    \toprule
    Heading1 & Heading2\tablefootnote{A
        ↪ footnote inside a table.} \\
    \midrule
    One     & Two
        ↪
        ↪ \\
    \bottomrule
  \end{tabular}
\end{table}
```

A table wider than the text runs into the margin. `\begin{adjustbox}{max width=\linewidth}` from the adjustbox package scales it down to the line width (the column width in a two-column layout), and only if it is too wide. Scaling shrinks the font, so for a table with long text in its cells, prefer wrapping the text with the column type `X` of the tabularx package, and for a very wide table, a landscape page (see `pdflscape`).

[6]A footnote inside a table.

Table V
WIDE TABLE SCALED TO THE LINE WIDTH

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1.11	2.22	3.33	4.44	5.55	6.66	7.77	8.88	9.99	10.10	11.11	12.12
Rain	Snow	Wind	Sun	Fog	Hail	Heat	Storm	Mist	Frost	Sleet	Calm

Table VI
TABLE WITH DIAGONAL LINE

Row \ Column	A	B
1	x	y

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
\begin{table}
  \caption{Wide table scaled to the line
      ↪ width}
  \label{tab:wide}
  \centering
  \begin{adjustbox}{max width=\linewidth}
    \begin{tabular}{lllllllllllll}
      \toprule
      Jan & Feb & Mar & Apr & May & Jun &
          ↪ Jul & Aug & Sep & Oct & Nov  &
          ↪ Dec  \\
      \midrule
      1.11 & 2.22 & 3.33 & 4.44 & 5.55 & 6.6
          ↪ 6 & 7.77 & 8.88 & 9.99 & 10.10 &
          ↪  11.11 & 12.12 \\
      Rain & Snow & Wind & Sun & Fog & Hail
          ↪ & Heat & Storm & Mist & Frost &
          ↪ Sleet & Calm \\
      \bottomrule
    \end{tabular}
  \end{adjustbox}
\end{table}
```

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
% Adapted from https://tex.stackexchange.
    ↪ com/a/468994/9075
\begin{table}
  \caption{Table with diagonal line}
  \label{tab:diag}
  \centering
  \begin{tabular}{|l|c|c|} % chktex 44
    \hline % chktex 44
    \diagbox{Row}{Column} & A & B \\
    \hline % chktex 44
    1                     & x & y \\
    \hline % chktex 44
  \end{tabular}
\end{table}
```

M. Source Code

Listing 1 shows source code written in XML. Line 2 contains a comment.

```
1 <listing name="example">
2   <!-- comment -->
```

```
<listing name="example">
  Floating
</listing>
```

Listing 2. Example XML listing – placed as floating figure

```
  <content>not interesting</content>
</listing>
```

Listing 1. Example XML Listing

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
% After a period, LaTeX sets the wider sentence space by default. The exception: a period after a capital letter
% is taken for an abbreviation or initial ("T. Scientist", "NASA") and gets the narrow inter-word space.
% "XML." ends a sentence, so \@ before the period restores the sentence space.
% (The other way round, "e.g.\ details" forces the narrow space after a lowercase abbreviation; ~ is a
% non-breaking inter-word space, as in "Prof.~Dr.~Eike Kraft".)
% An abbreviation set with \gls{xml} needs no \@: glossaries ends it with a non-letter, so the period after it
% gets the sentence space. The plain "XML" here is for documents without the glossaries package.
\Zcref{lst:XML} shows source code written in XML\@.
\Zcref{line:comment} contains a comment.

\begin{lstlisting}[
  language=XML,
  caption={Example XML Listing},
  label={lst:XML}]
<listing name="example">
  <!-- comment --> (* \label{line:comment} *)
  <content>not interesting</content>
</listing>
\end{lstlisting}
```

One can also add `float` as parameter to have the listing floating. Listing 2 shows the floating listing.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{lstlisting}[
  % one can adjust spacing here if required
  % aboveskip=2.5\baselineskip,
  % belowskip=-.8\baselineskip,
  float,
  language=XML,
  caption={Example XML listing -- placed as floating figure},
  label={lst:flXML}]
<listing name="example">
  Floating
</listing>
\end{lstlisting}
```

```
{
  key: "value"
}
```

Listing 3. Example JSON listing – placed as floating figure

```
public class Hello {
    public static void main (String[] args) {
        System.out.println("Hello World!");
    }
}
```

Listing 4. Example Java listing

One can also typeset JSON as shown in Listing 3.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{lstlisting}[
  float,
  language=json,
  caption={Example JSON listing -- placed as floating figure},
  label={lst:json}]
{
  key: "value"
}
\end{lstlisting}
```

Java is also possible as shown in Listing 4.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{lstlisting}[
  caption={Example Java listing},
  label=lst:java,
  language=Java,
  float]
public class Hello {
    public static void main (String[] args) {
        System.out.println("Hello World!");
    }
}
\end{lstlisting}
```

N. Itemization

One can list items as follows:

- Item One
- Item Two

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{itemize}
  \item Item One
  \item Item Two
\end{itemize}
```

With the package paralist, one can create itemizations with lesser spacing:

- Item One
- Item Two

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{compactitem}
  \item Item One
  \item Item Two
\end{compactitem}
```

One can enumerate items as follows:

1) Item One
2) Item Two

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{enumerate}
  \item Item One
  \item Item Two
\end{enumerate}
```

With the package paralist, one can create enumerations with lesser spacing:

1) Item One
2) Item Two

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{compactenum}
  \item Item One
  \item Item Two
\end{compactenum}
```

With paralist, one can even have all items typeset after each other and have them clean in the TeX document:

1) The first item, 2) the second item, and 3) the third item appear in one line.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
\begin{inparaenum}
  \item The first item,
  \item the second item, and
  \item the third item appear in one line.
\end{inparaenum}
```

Right	Left	Centered
12	12	12
123	123	123

Table VII
A TABLE WRITTEN IN MARKDOWN

O. *Writing in Markdown*

If you are new to LaTeX, you can start by writing your text in Markdown using the markdown package. LaTeX then takes care of the typesetting. Markdown and LaTeX can be mixed: Table VII is written in Markdown and referenced from LaTeX. A whole section can be written in a separate Markdown file included by `\markdownInput`, as Section III shows. The markdown-latex-quickstart shows a complete document written in Markdown.

Text can be *emphasized* or **bold**, and `code` is set in a monospaced font. A citation looks like this [3], and here is a footnote.[7] Referenced footnotes work as well[8].

- first point
- second point
  - sub-point

1. firstly
2. secondly

Table VII shows a table with caption. Its identifier `{#tbl:markdown}` is referenced by `<#tbl:markdown>`.

Corresponding LaTeX code of paper-basic-listings.tex

```
œŒűŰőŐłŁıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„"",''
  \begin{markdown}
Text can be *emphasized* or **bold**, and `
    ↪ code` is set in a monospaced font.
A citation looks like this [@mwe], and here
    ↪  is a footnote.^[This is an inline
    ↪ footnote.]
Referenced footnotes work as well[^detail].

[^detail]: This is a footnote defined at a
    ↪ different place in the text.

- first point
- second point
  - sub-point

1) firstly
2) secondly

| Right | Left | Centered |
|------:|:-----|:--------:|
```

[7]This is an inline footnote.
[8]This is a footnote defined at a different place in the text.

```
|    12 | 12  |    12    |
|   123 | 123 |   123    |

: A table written in Markdown {#tbl:
    ↪ markdown}

<#tbl:markdown> shows a table with caption.
Its identifier `{#tbl:markdown}` is
    ↪ referenced by `<#tbl:markdown>`.
\end{markdown}
```

P. Other Features

The words “workflow” and “dwarflike” can be copied from the PDF and pasted to a text file.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒŭŐőŁłıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
The words \enquote{workflow} and \enquote{
    ↪ dwarflike} can be copied from the PDF
    ↪  and pasted to a text file.
```

The symbol for powerset is now correct: $\wp$ and not a Weierstrass p ($\wp$).
$\wp(1,2,3)$

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒŭŐőŁłıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
The symbol for powerset is now correct: $\
    ↪ powerset$ and not a Weierstrass p ($\
    ↪ wp$).

$\powerset({1,2,3})$
```

Brackets work as designed: <test> One can also input backticks in verbatim text: `` `test` ``.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒŭŐőŁłıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
Brackets work as designed:
<test>
One can also input backticks in verbatim
    ↪ text: \verb|`test`|.
```

Acronyms can be typeset in running text with `\initialism`, which sets them as small caps that blend into the surrounding lowercase text. It is the lightweight alternative to the full acronym and glossary management (`\gls`, with an automatic list of abbreviations) provided by the thesis variants: `\initialism` only typesets the acronym and does not add it to any list. The macro and the ready-made acronyms (`\OMG`, `\BPEL`, `\BPMN`, `\UML`) are defined in `commands.tex`, where you can add your own.

The UML is maintained by the OMG; related standards include BPEL and BPMN. Any acronym can be set inline with REST or HTTP.

Corresponding LaTeX code of `paper-basic-listings.tex`

```
œŒŭŐőŁłıčČšŠžŽřŘěĚůŮşŞğĞąĄęĘńŃśŚźŹżŻ€…--„“”,‘’
The \UML is maintained by the \OMG; related
    ↪  standards include \BPEL and \BPMN.
Any acronym can be set inline with \
    ↪ initialism{REST} or \initialism{HTTP
    ↪ }.
```

V. Conclusion and Outlook

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Acknowledgment

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which comes right before the reference section in your document [4].

References

[1] O. Kopp *et al.*, “Winery – A Modeling Tool for TOSCA-based Cloud Applications,” in *Proceedings of 11th International Conference on Service-Oriented Computing (ICSOC’13)*, ser. LNCS, vol. 8274. Springer Berlin Heidelberg, 2013, pp. 700–704, doi: 10.1007/978-3-642-45005-1_64.

[2] T. Binz, G. Breiter, F. Leymann, and T. Spatzier, “Portable Cloud Services Using TOSCA,” *IEEE Internet Computing*, vol. 16, no. 03, pp. 80–85, May 2012, doi: 10.1109/mic.2012.43.

[3] M. Scharrer, *The mwe Package*, 2017. [Online]. Available: http://texdoc.net/mwe

[4] B. Veytsman, “Latex class for the association for computing machinery – acknowledgement information,” Aug. 2021. [Online]. Available: https://github.com/borisveytsman/acmart/blob/1704c8bf7eee92a1515ff755f5118b6a22bb1f8e/samples/samples.dtx#L709

All links were last followed on October 5, 2020.